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GROUP D

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1. Objective

This report applies regression and clustering on the "Airbnb-NYC-2019" (Dgomonov, 2019) dataset to answer: "Can Airbnb identify pricing patterns and listing segments to support hosts with competitive pricing and segment guidance?"

2. Methodology

CRISP-DM links the machine-learning workflow with the business question, allowing loop-back for adjustment (Chapman et al., 2000; Schröer, Kruse and Gómez, 2021).

EDA, data cleaning and feature preparation made the dataset suitable for modelling by handling missing, zero, no-review and extreme values that distorted results. Price was transformed into "log_price" as log transformation is widely used for skewed variables (Feng et al., 2014); categorical variables were binary-encoded, and clustering inputs standardised, as regression requires numerical inputs and K-Means is scale-sensitive (Bishop, 2006; James et al., 2023).

For pricing prediction, three regression models were compared in a nested progression from simple to complex following Bishop (2006) and James et al. (2023) model-comparison guidance with predictors that improve performance while keeping interpretation by using observable listing signals.

Model-1 used room-type dummy variables as baseline. Model-2 added nine selected features. Model-3 added degree-2 terms to test whether limited extra complexity improves practical prediction. An 80/20 hold-out split was used with R^2 and RMSE compared on the 20% test set using "log_price".

Bishop (2006) presents K-Means as an unsupervised method that groups observations around cluster centres on a distance-based objective, and the scikit-learn K-Means documentation also describes the method through inertia and within-cluster sum of squares (Pedregosa et al., 2011; scikit-learn, no date). Therefore, K-Means was used on scaled listing features to prevent scale distortions. K-values from 2 to 6 were tested using elbow/SSE and silhouette logic (Rousseeuw, 1987; Thinsungnoen et al., 2015), with K=3 selected as the practical link between segmentation and business interpretation. Inputs and diagnostics are in Appendix.

3. Limitations

The dataset is a 2019 snapshot lacking customer data, guests or bookings. Availability lacks underlying reasons and is treated cautiously as a listing signal rather than a customer-demand proxy, hindering direct customer segmentation. Extended attributes (Airbnb, 2023) and photos (Ert, Fleischer and Magen, 2016) influence pricing and trust in renting business. Lacking these features, alongside facilities, ratings, and seasonality, renders the price prediction model incomplete. Lacking valid customer proxies, models prioritised interpretable, decision-led guidance over complex models requiring unavailable demand data. Thus, limitations shift segmentation to listing level and price prediction to listed-price guidance. Future work should add missing price driver features for a pricing engine, or alternatively, if dataset remains unchanged, complement current modelling with the addition of a Random Forest price-band classification for a richer price guidance. This was excluded due to length, time and resource constraints.

4. Results and analysis

4.1 Market insights

After cleaning, the dataset has 48.847 listings with strongly skewed prices: median $106, mean $153, and 99th percentile $799 with outliers distorting the curve upward. Manhattan and Brooklyn concentrate 85% of listings, census data show they account for 50.3% and 30.6% of population and land respectively (United States Census Bureau, nd.)

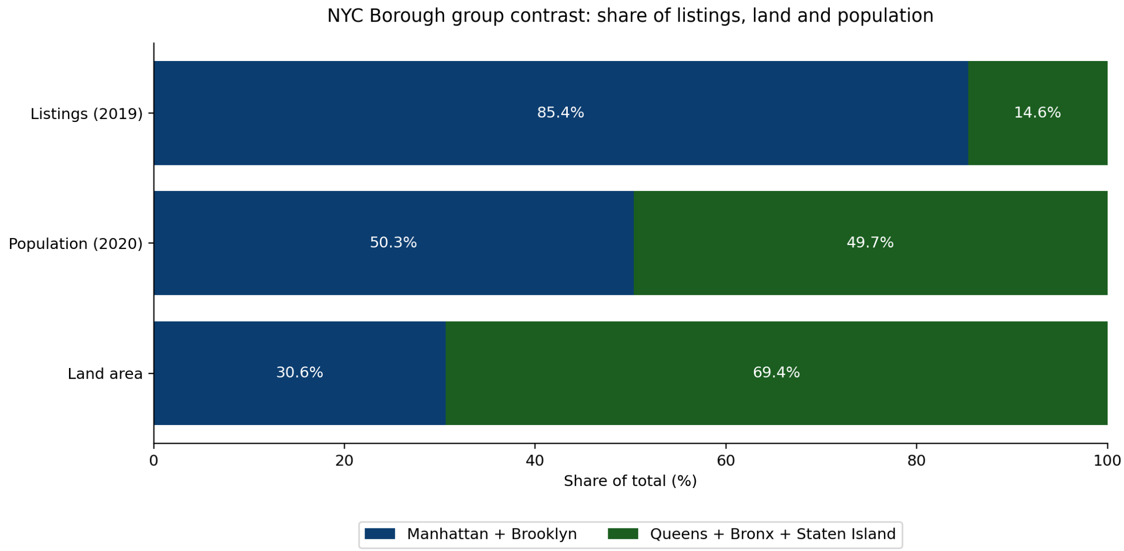

Figure 1: borough group contrast

Availability: 35.8% of listings are unavailable for unknown reasons, this could mean high-demand, or inactivity/withdrawn, thus it is a cautious listing signal, not demand.

4.2 Regression: price prediction and main price signals

Model 3 performs best, explaining 54.4% of observed “log_price” variation. Table 1 shows the model comparison.

Table 1: Regression model comparison

Model	Tested	R^2	RMSE	Interpretation
Model 1: Baseline	Room type	0.3791	0.5444	Explains meaningful variation.
Model 2: Multiple linear	Listing, location, review, availability and host-scale features	0.4946	0.4912	More features, prediction improves.
Model 3: Polynomial degree 2	Added limited complexity	0.5436	0.4667	Best, explains 54.4% of observed "log_price".

Table 2 summarises pricing evidence from EDA and regression:

Table 2: Pricing signals evaluation

Pricing signal	Outcome	Business interpretation
Room type	Strongest tested signal; baseline R^2 = 0.3791; top two Model 2 coefficients are room-type dummies	First peer comparison should be by room-type.
Borough/location	Manhattan + Brooklyn hold 85% of listings	NYC is not one flat market; location matters.
Availability	35.8% zero availability; feature helps model	Useful signal, but reason unknown.
Reviews / activity	Review variables included as tested features	Activity/reputation context, not bookings.

Results show that Model 3 is the best pricing guide, while Model 2 identifies the underlying observable signals for the benchmark. The evidence supports room-type peer benchmarking but remains associative, not causal. Coefficient details are in Appendix.

4.3 Market segmentation

Because the dataset contains listing attributes but not customer ones, and room-type was the strongest observed price signal, K-Means was used to identify listing peer groups based on room-type led segments, summarised in Table 3:

Table 3: K-means listing segments

Segment	Share	Price	Availability	Borough skew
Entire home	**52.6%**	**$176**	113 days/annum	Manhattan-skewed
Private room	**45.1%**	**$72**	110 days/annum	Brooklyn-skewed
Shared room	**2.4%**	**$54**	162 days/annum	Mixed

Figure 2 shows cluster visualisation:

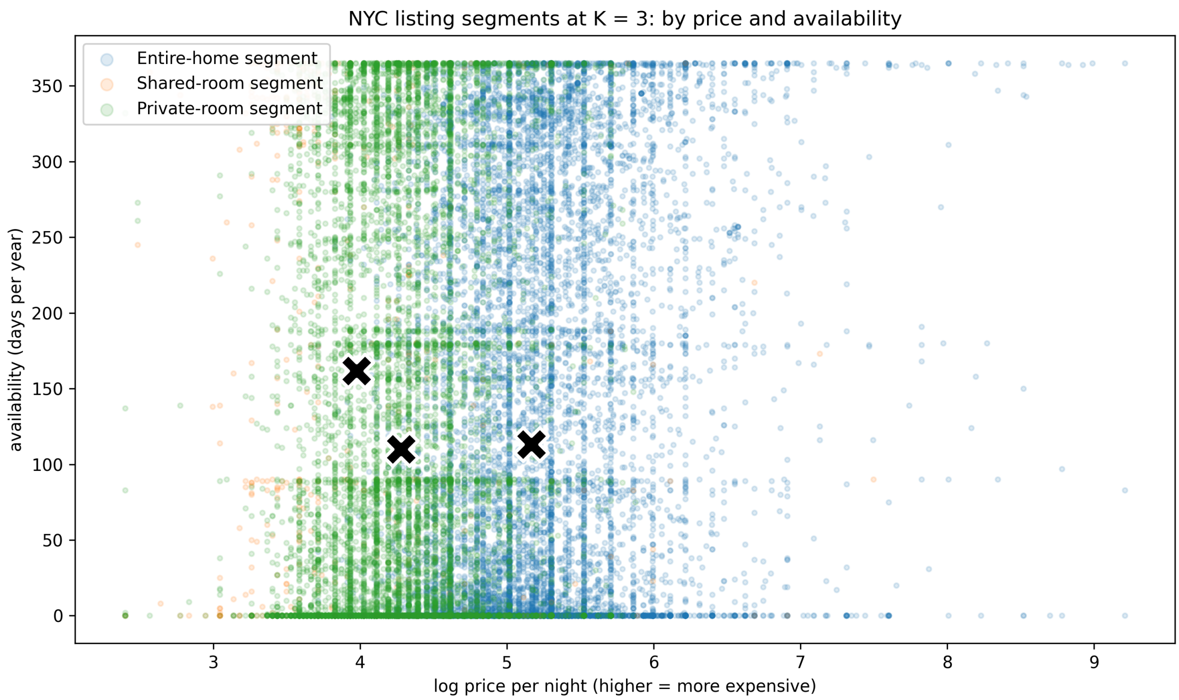

Figure 2: listing room-type and log_price

The X markers show cluster centres; higher “log_price” means more expensive listings. The overlap supports segment use as broad groups for peer comparison and host guidance, not hard boundaries.

5. Business recommendations

5.1 Differentiation leveraging peer-price and segmentation

Porter (1980) links better business outcomes to differentiation and focus; Airbnb should test Model 3 as a peer-price benchmark, Model 2 to explain observable signals, and apply cluster segments to customise host-guidance: entire-home hosts receive premium-positioning advice, private-room hosts receive value and occupancy guidance, and shared rooms are treated as niche. In this way, Model 3 provides the benchmark and K-Means defines the peer group, avoiding treating NYC as a flat market.

5.2 Zero availability SWOT

Unavailability could represent an opportunity or risk depending on the reason behind; and SWOT classification can unearth action benefits. Fully booked listings could indicate an opportunity to work on pricing review, whereas unavailable ones represent a risk and potential cost saving; identifying the reason and acting on it could deliver business value matching opportunities and threats with conditions (Weihrich, 1982; Helms and Nixon, 2010).

5.3 Market protection

Manhattan and Brooklyn listing concentration suggest a red-ocean market, whereas Queens, the Bronx and Staten Island, with lower listing shares despite comparable populations and more land, could represent a blue-ocean market as depicted by Kim and Mauborgne (2005). Airbnb could test wetter these markets belong to those categories by implementing room-type peer benchmarks in concentrated areas to protect revenue, while piloting new hosts recruitment in lower-share ones to potentially increase revenue.

In conclusion, limitations constrain more robust price prediction and customer segmentation; however, the results provide pricing guidance, listing segmentation, and testing of several initiatives that could lead to better business outcomes.

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7. Appendix

Pricing signal design

First model, baseline, room-type dummy variables; second model with selected nine features, the same room-type dummies with neighbourhood-group dummy variables and numeric listing features such as latitude, longitude, minimum_nights, number_of_reviews, reviews_per_month, availability_365 and calculated_host_listings_count; third model, degree 2 polynomial terms on the same feature base. Degree 2 was used to test simple curved patterns and feature-pair effects without moving into more complex models that are harder connect with business indicators and analysis.

Table 4: Largest absolute coefficients from Model 2

Rank	Signal from Model 2	Coefficient
1	room_type_Private room	-0.3795
2	room_type_Shared room	-0.1782
3	longitude	-0.1418
4	neighbourhood_group_Manhattan	0.1397
5	availability_365	0.1151

Coefficients are model evidence, not causal effects. Entire home/apt is the room-type reference group, so the negative private-room and shared-room coefficients show lower listed-price position compared with that reference.

K-Means inputs and diagnostics

For clustering, the input features were log_price, minimum_nights, reviews_per_month, availability_365 and room-type dummy columns. The original room_type field was converted into dummy yes/no columns with drop_first=True, so K-Means could use room type without treating it as an ordered number. K values were tested from K = 2 to K = 6 using elbow and silhouette logic (Rousseeuw, 1987; Thinsungnoen et al., 2015). The final groups align mostly with room type, this was expected because room type is both part of the clustering input and the strongest observed price signal, confirming the richness in listing related features.

Figure 3 assesses the selected K value using listing-level silhouette scores. Positive medians support the three-way split. Nevertheless, some overlap remains, so the segments are used for assistance or guiding but cannot be used as definitive groups (Rousseeuw, 1987).

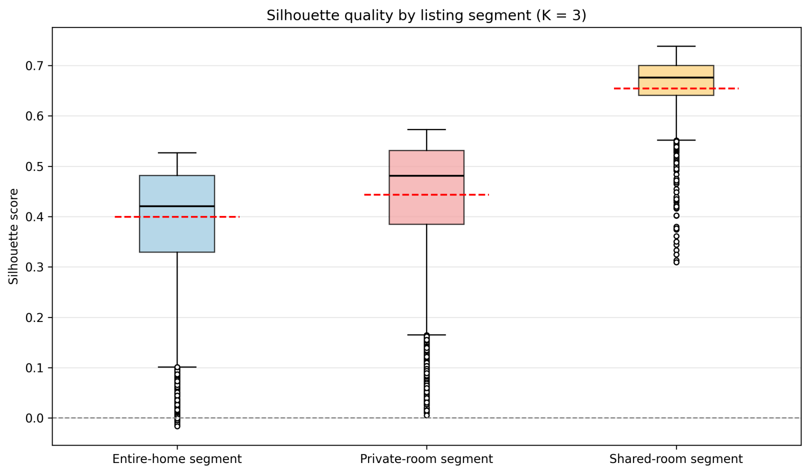

Figure 3: Silhouette quality by listing segment